DATS 6102 - Final Group Project

NoSQL Database Research, Implementation & Evaluation | Fall 2026 | 200 Points (20% of Course Grade)

I am providing this class-wide project guide as the primary source for all common requirements, deadlines, grading, final report, presentation, demo, peer review, and AI expectations. Each group will also receive a short database-specific supplement that contains only the requirements that differ for that assigned database. Read this overview first, then use your group supplement for the database-specific technical requirements.

Project Purpose

I am assigning each group a NoSQL database that is not part of the regular hands-on database coverage in this course. I expect each group to study how its assigned technology works, build a small working solution, compare it directly with MySQL, compare it with MongoDB or Neo4j, and present a technical demonstration.

Group Assignments

Group	Assigned Database	Primary Model / Focus	Group Size	Group Members
1	Apache CouchDB	Document database; JSON/HTTP/replication	4 students	Damian Mendoza; Ingri Quevedo; Stephanie Armstrong; Hassan Saleh H Alshamrani
2	Redis	Key-value / in-memory data structures	5 students	Kaitlyn Jones; Abdulaziz Alsharidi; Simon Tao; Vik Becker; Anton Mcdonald
3	Couchbase Community Edition	Document + key-value	5 students	Mak Trnka; Galileo Frank Perricone; Constantine Zhang; Arpita Gonnade; Alexei Kiliakov
4	ArangoDB Community Edition	Multi-model: document + graph	5 students	Abdulrahman Muqahhis; Mia Alecia Goins; Ella Smith; Menna Yohannes; Hugo Ndizeye

Common Requirements for All Groups

- Select a realistic business or technical problem and a dataset appropriate for the assigned NoSQL platform. I must approve your proposal before you proceed with full implementation.
- Use public, synthetic, or data I have approved. Do not use confidential, restricted, or personally sensitive data.
- Install/configure the assigned free/community/open-source database locally or through another no-cost option that I approve. Students are not expected to purchase software or cloud services.
- Design an appropriate NoSQL data model for the selected scenario and explain the design decisions.
- Load enough data to demonstrate meaningful modeling and querying. As a general guideline, target at least 500 records/documents/items unless the platform or use case justifies a different scale.
- Demonstrate basic create/read/update/delete operations, or the closest platform-equivalent operations where traditional CRUD does not apply exactly.
- Demonstrate at least four meaningful queries/operations that answer business or technical questions.
- Demonstrate at least one capability that is distinctive or especially important for the assigned database.
- Compare the assigned NoSQL database directly with MySQL and explain whether the same business problem could reasonably be implemented in MySQL, including advantages and tradeoffs.
- Compare the assigned platform with at least one NoSQL database covered in class (MongoDB or Neo4j).
- Every member must participate in the final presentation and understand the overall project. I may ask any group member individual technical questions.
- Follow the course AI policy. Any permitted AI assistance must be disclosed and validated by the group.

Project Timeline

Date / Week	Milestone
Week 9 - Thu, Oct. 22	Project released; initial group meeting in class.
Mon, Oct. 26 @ 10:00 PM	Part 1 - Project Proposal due before Week 10 class.
Week 10 - Thu, Oct. 29	Proposal review / project planning.
Mon, Nov. 9 @ 10:00 PM	Part 2 - Technical Progress Submission due.
Week 12 - Thu, Nov. 12	In-class technical check-in. Each group must show concrete working progress.
Mon, Nov. 30 @ 10:00 PM	Two final group submissions due separately: (1) Final Technical Report and supporting implementation evidence; (2) PowerPoint Presentation. Individual peer review is also due separately.
Week 15 - Thu, Dec. 3	Final presentations and technical demonstrations in class.

Deliverables

Part 1 - Project Proposal (20 points)

Due Monday, October 26, 2026, at 10:00 PM. Recommended length: 2-3 pages plus one preliminary architecture/data-model diagram.

- Group number and names of all group members.
- Assigned NoSQL database and its primary database model.
- Business problem, intended users, and why the problem is suitable for the assigned NoSQL platform.
- Dataset source, format, approximate size, and planned preparation/cleaning.
- Preliminary NoSQL data model or structure.
- Preliminary architecture/data-flow diagram.
- Short preliminary comparison with MySQL: why might this scenario benefit from the assigned NoSQL database?
- Responsibility matrix showing each member's primary responsibility.
- At least two technical risks, setup questions, or issues the group expects to investigate.

Part 2 - Technical Progress Check-In (20 points)

Submission due Monday, November 9, 2026, at 10:00 PM. Recommended length: 2-3 pages plus screenshots, query/output examples, or other technical evidence as needed.

Written/technical submission due Monday, November 9, 2026, at 10:00 PM; in-class check-in Thursday, November 12.

- Database environment installed/configured or otherwise accessible.
- Initial data structures created and sample/project data loaded.
- At least two meaningful queries/operations working.
- Updated data model/architecture diagram.
- Evidence of progress on the MySQL comparison.
- Status of each member's assigned responsibility.
- Remaining technical issues, risks, or changes from the proposal.

Final Submission and Presentation

I am requiring two separate final group submissions in Blackboard. Both are due Monday, November 30, 2026, at 10:00 PM. The live technical demonstration will be graded during the December 3 presentation.

Submission 1 - Final Technical Report (80 points / 40%)

- Recommended length: 8-12 pages, excluding the title page, references, screenshots, and appendices. This is where I expect the detailed technical analysis, implementation discussion, MySQL comparison, NoSQL comparison, architecture evaluation, evidence, and conclusions.
- Include final architecture/data-model diagram(s), implementation evidence, query/command examples or screenshots, dataset/source information, contribution statement, references, and the required AI-use disclosure if applicable.

Submission 2 - PowerPoint Presentation (40 points / 20%)

- Submit the PowerPoint file separately. The formal presentation should be approximately 15 minutes and include at least 10 substantive content slides, not counting the title slide or references slide. I will grade the written slides and the group's oral communication separately from the live demo.

Live Technical Demonstration (20 points / 10%)

- Plan approximately 8-10 minutes for the live demonstration. I expect the group to show the working database, loaded data, meaningful operations/queries, and at least one database-specific capability. This component is graded live and does not require a third final group upload.

Questions and Timing

- After the formal presentation and demo, plan approximately 5-10 minutes for questions from me. Total time per group should be approximately 30-35 minutes. Every group member must speak and contribute.

Peer Review (20 points / 10%)

- Each student must submit the confidential peer evaluation individually. This is separate from the two final group submissions.

Presentation Expectation

Final presentation timing: plan approximately 15 minutes for the formal PowerPoint presentation, followed by 8-10 minutes for the live technical demonstration and approximately 5-10 minutes for questions from me. The total time for each group should be approximately 30-35 minutes. Every group member must speak and contribute. The formal presentation must include at least 10 substantive content slides, not counting the title slide or references slide. The PowerPoint presentation (40 points) and live technical demonstration (20 points) are graded as separate components.

Grading Summary - 200 Points

Component	Points	What I will evaluate
Part 1 - Project Proposal	20	Business problem, dataset, preliminary model, team roles, relational/MySQL rationale, risks.
Part 2 - Technical Check-In	20	Working environment, loaded sample data, early operations/queries, updated design, team status.
Final Technical Report	80	Detailed technical analysis, design/implementation documentation, MySQL comparison, NoSQL comparison, architecture/technology evaluation, evidence, conclusions, and references.
PowerPoint Presentation & Communication	40	Quality of written slides, organization, technical accuracy, visual clarity, oral delivery, balanced participation, and timing.
Live Technical Demonstration	20	Working database, data loading, meaningful operations/queries, platform-specific capability, technical correctness, and ability to explain what is shown.
Peer Review - Individual	20	Confidential teammate evaluation scored individually.

Final Technical Report Breakdown - 80 Points (40%)

Report Area	Points	What I expect
NoSQL Technical Analysis	20	Architecture, data model, consistency/transactions, indexing/querying, scaling/availability, deployment/licensing, strengths, limitations, and appropriate use cases.

Data Model, Implementation & Results	20	Project-specific design, data preparation/loading, implementation choices, meaningful operations/queries, results, screenshots/evidence, and explanation of technical decisions.
Relational Database / MySQL Comparison	20	Scenario-specific relational mapping, tables/keys/relationships, joins, schema, transactions/consistency, indexes, scaling, tradeoffs, and a defensible recommendation.
Architecture Evaluation & NoSQL Comparison	15	Why the assigned platform fits or does not fit the workload, risks/alternatives, and comparison with MongoDB or Neo4j.
Report Quality, Evidence & References	5	Professional organization, clarity, credible sources, correct citations, labeled figures/evidence, and a clear conclusion.

The report is the primary location for detailed analysis. I do not expect the PowerPoint slides to repeat every detail from the report.

Relational Database / MySQL Comparison - Required for Every Group

The comparison must be specific to the group's selected dataset and business scenario, not just a generic "SQL vs. NoSQL" discussion. Address all of the following:

- How the NoSQL data model maps to MySQL tables, rows, columns, primary keys, and foreign keys.
- Schema approach: fixed relational schema versus the assigned database's schema behavior.
- How relationships are represented versus joins and foreign keys.
- How querying and data manipulation differ from SQL SELECT/INSERT/UPDATE/DELETE.
- Transactions, consistency, concurrency, and durability compared with MySQL.
- Indexes and query-performance considerations compared with MySQL.
- Horizontal/vertical scaling and availability differences.
- Which parts of the business problem would be easier or harder in MySQL.
- A final recommendation: MySQL, the assigned NoSQL database, or a combined architecture - and why.

Generative AI and Academic Integrity

The course AI policy applies to every part of this project. I expect the final work to demonstrate your group's own technical understanding, design decisions, implementation work, and ability to explain and defend the assigned database.

- Permitted uses of generative AI include limited assistance with brainstorming an initial approach, explaining a concept you are trying to understand, debugging a limited portion of work your group has already attempted, and improving grammar, organization, or formatting of material your group originally wrote.
- Generative AI may not be used to generate the complete project, report, presentation, technical analysis, data model, schema, full set of queries/commands, configuration, or other required project solution for submission.
- Do not copy or lightly paraphrase AI-generated technical content and submit it as your group's work. AI-generated claims, commands, citations, and recommendations must be independently verified using course materials, official documentation, testing, or other credible sources.
- If any group member uses a generative AI tool at any stage of the project, the final report must include a short Generative AI Use Disclosure identifying the tool, how it was used, what portion of the work it affected, and how the group verified and revised the output.
- Generative AI may not be used during the live presentation or demonstration to answer questions that group members are expected to understand themselves. I may ask any member questions about submitted work or technical decisions.
- Submitting substantially AI-generated work as your own, fabricating technical evidence or references, or failing to disclose AI assistance may be treated as an academic integrity issue under the course and university policies.

Graduate-Level Expectation

I will evaluate more than whether your database can store and query data. I expect your group to explain why the chosen data model fits the workload, demonstrate a working implementation, identify tradeoffs, support technical claims with credible documentation, and show that every member understands the design.

PowerPoint Presentation & Communication Breakdown - 40 Points (20%)

Presentation Area	Points	What I expect
Content Selection & Organization	15	A focused 15-minute presentation that communicates the problem, database, design, major findings, comparisons, and recommendation in a logical sequence.
Technical Accuracy & Understanding	10	Accurate explanations, appropriate terminology, and clear evidence that presenters understand the work.
Visual Quality of Slides	5	Readable, professional, informative slides and visuals; at least 10 substantive content slides excluding title and references.
Oral Communication & Delivery	5	Clear, professional delivery rather than reading directly from slides.
Balanced Participation & Timing	5	All members contribute meaningfully and the formal presentation stays near the 15-minute target.

Live Technical Demonstration Breakdown - 20 Points (10%)

Demo Area	Points	What I expect
Working Environment & Loaded Data	5	The assigned database is running, and the project data is visibly loaded/available.
Meaningful Operations / Queries	8	The group demonstrates required CRUD/platform-equivalent operations and meaningful business/technical queries successfully.
Database-Specific Capability	4	The group demonstrates and explains at least one capability distinctive or especially important to the assigned platform.
Technical Explanation & Demo Readiness	3	The demo is organized, presenters explain what is happening, and the group can recover or explain clearly if a minor issue occurs.